

Sd Rationality Check

Symptoms

Dual Fuel operation will stop. Engine will continue to operate in diesel **only** mode.

How To Use This Tree

Circuit Description:

Alarm is activated if difference between engine power based on diesel consumption and engine power based on value transmitted from OEM/ECM is greater than 25 percent of engine rated power at given rpm and altitude for at least 4s prior to start of the gas substitution.

Conditions for Running the Diagnostics:

Only active in MAN and AUT before transition to Dual Fuel operation, when all condition for running in Dual fuel are met.

Conditions for Activating the Fault Message:

Alarm is activated if difference between engine power based on diesel consumption and engine power based on value transmitted from OEM/ECM is greater than 25 percent of engine rated power at given rpm and altitude for at least 4s prior to start of the gas substitution.

Conditions for Clearing the Fault Code Automatically:

None.

Conditions for Clearing the Fault Code Manually:

Latching fault, reset is required. Once Sd message is activated controller switches to Diesel only state, and the message will become inactive. When problem is solved it can be cleared from IMON or IV5 by pressing fault reset.

Shoptalk

Possible causes include:

- Incorrect power reading
- Incorrect fuel consumption reading due to injector drift
- Engine fuel system failure.

Fault Message FLS Detonation Analog 4-20 mA Signal is Missing from IBF CU Input at Analog Input Channel		
Code or Message	Reason	Effect
Sd Rationality Check	Rationality Check indicates the calculated diesel power (kW) is out of tolerance.	Dual fuel control system will not allow gas operations. Dual fuel controller will stop gas flow.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check for errors.	
	STEP 1A. Check the engine ECM for engine fault codes.	Engine ECM indicates fault codes?
	STEP 1B. Review the fault message(s).	Sd RationalCheck alarm code present?
	STEP 1C. Review the fault message(s).	Alarm Codes other than Sd RationalCheck present?
STEP 2.	Engine load not within set limit.	
	STEP 2A. Verify source for engine power value.	Pump using Real Power message shown in J1939?
	STEP 2B. Verify Real Power transmitted value (J1939).	Do J1939 transmitted power limits correspond with pump power rating?
	STEP 2C. Verify engine load limits (Converted).	Do power limits correspond with pump power rating?
STEP 3.	Check for fault messages.	
	STEP 3A. Reset the fault.	Fault returns?

STEP 1. Check for errors.

STEP 1A. Check the engine ECM for engine fault codes.

Conditions :

- Turn keyswitch ON.
- Connect the recommended Cummins® electronic service tool or equivalent.

Action	Specification/Repair	Next Step
Use the recommended Cummins® electronic service tool or equivalent to read the fault code.	Engine ECM indicates fault codes? YES	Troubleshoot engine fault codes prior to Troubleshooting Alarm Codes.
	Engine ECM indicates fault codes? NO	1B

STEP 1B. Review the fault message(s).

Conditions :

- Power ON dual fuel control module.
- Connect IntelliMonitor to the dual fuel control panel.

Action	Specification/Repair	Next Step
Check local display panel or IntelliMonitor for fault messages in alarm list and history.	Sd RationalCheck alarm code present? YES	1C
	Sd RationalCheck alarm code present? NO	Return the pump to service and monitor.

STEP 1C. Review the fault message(s).

Conditions :

- Power ON dual fuel control module.
- Connect IntelliMonitor to the dual fuel control panel.

Action	Specification/Repair	Next Step
Check local display panel or IntelliMonitor for fault messages in alarm list and history.	Alarm Codes other than Sd RationalCheck present? YES	Troubleshoot all other error codes prior to troubleshooting Sd RationalCheck.
	Alarm Codes other than Sd RationalCheck present? NO	2A

STEP 2. Engine load not within set limit.

STEP 2A. Verify source for engine power value.

Conditions :

- Power ON dual fuel control module.
- Connect InteliMonitor to the dual fuel control panel.

Action	Specification/Repair	Next Step
Verify power source in PLC Monitor. (Converted or J1939).	Pump using Real Power message shown in J1939? YES	2B
	Pump using Real Power message shown in J1939? NO	2C

STEP 2B. Verify Real Power transmitted value (J1939).

Conditions :

- Power ON dual fuel control module.
- Connect InteliMonitor to the dual fuel control panel.
- Connect the recommended Cummins® electronic service tool or equivalent

Action	Specification/Repair	Next Step
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Monitor value for Real Power transmitted via J1939 on IV5 screen 7 OR IMON – ECU – Pwr-Real R. Verify engine is operating within power limits. (2250 HP engine 373-1578 kW) (2500 HP engine: 373-1748 kW) If both 4-20ma input and J1939 message are present, the J1939 message takes priority and control ignores the converted signal. All limits should still be in the same range.	Do J1939 transmitted power limits correspond with pump power rating? YES	3A
	Do J1939 transmitted power limits correspond with pump power rating? NO Repair : For pump with J1939 Real Power Value: Connect the recommended Cummins® electronic service tool or equivalent and verify that power rating transmitted through the ECM is correct. If value is incorrect from the OEM, then the OEM or customer needs to correct this value.	Contact a Cummins® Authorized Repair Location.

STEP 2C. Verify engine load limits (Converted).

Conditions :

- Power ON dual fuel control module.
- Connect InteliMonitor to the dual fuel control panel.
- Connect the recommended Cummins® electronic service tool or equivalent.

Action	Specification/Repair	Next Step
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Compare the value the OEM is transmitting for Engine HP matches value read in IMON -Analog CU - FracPumpHP OEM sends linear signal as 4-20 mA corresponding to 0-2500 HP (based on engine rating).	Do power limits correspond with pump power rating? YES	3A
	Do power limits correspond with pump power rating? NO Repair : For pumps with converted power limits: Using a DMM ensure that power rating signal on BF1-A2 is correct. If value is incorrect from the OEM, then the OEM or customer needs to correct this value. If value from OEM is correct and error persists replace IBF Module.	Repair complete

STEP 3. Check for fault messages.

STEP 3A. Reset the fault.

Conditions :

- Engine **not** operating.

Action	Specification/Repair	Next Step

- Reset the fault on the control panel or through IntelliMonitor. - Operate the engine under conditions to allow gas substitution.	Fault returns? YES Repair : Verify all engine systems are working correctly. Correct any malfunctions on the engine. Attempt to reset the fault. Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	1A
	Fault returns? NO	Repair complete.